

1. Administrative & Study Identification

Full Study Title: A Phase 1 Clinical Study to Investigate the Safety and Pharmacokinetics of MK-5684 in China Participants With Metastatic Castration-Resistant Prostate Cancer **Short Title/ Acronym:** A Study of Opevesostat (MK-5684) in China Participants With mCRPC (MK-5684-001) **Protocol Number:** MK-5684-001 **NCT Number:** NCT06136598 **Posting ID:** 892317133212947536 **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Opevesostat (MK-5684) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of opevesostat in the treatment of male Chinese participants with metastatic castration-resistant prostate cancer (mCRPC) and to characterize the pharmacokinetic profile of opevesostat. There are no formal hypotheses to be tested in this study. **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Duration of Response (DOR), Overall Survival (OS), Radiographic Progression-Free Survival (rPFS), and Prostate-specific antigen (PSA) response rate.

2.2 Background & Rationale To address the clinical need for new therapeutic options in patients with metastatic castration-resistant prostate cancer (mCRPC) who have progressed despite existing hormone therapies and chemotherapy. Opevesostat (MK-5684) is a novel CYP11A1 inhibitor designed to manage the disease. This Phase 1 study focuses on gathering critical information about the drug's safety, dose-limiting toxicities, and pharmacokinetics (absorption, distribution, elimination) within the Chinese patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

3.2 Planned Enrollment & Duration Target \$N\$: 14 participants **Number of Sites:** Multiple locations in China **Estimated Study Start:** 30-Jan-2024 **Estimated Primary Completion:** 09-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histologically or cytologically confirmed adenocarcinoma of the prostate without small cell histology. 2. Has prostate cancer while receiving androgen deprivation therapy (ADT), or post-bilateral orchiectomy, within 6 months before screening. 3. Has evidence of progression >4 weeks since last flutamide treatment or >6 weeks since last bicalutamide or nilutamide treatment. 4. Has evidence of metastatic disease documented by either bone lesions on bone scan and/or soft tissue shown by CT/MRI. 5. Has disease that progressed during or after treatment with at least 1 line of next-generation hormonal agents (NHAs) for hormone-sensitive prostate cancer (HSPC) or castration-resistant prostate cancer (CRPC) for at least 8 weeks (at least 14 weeks for participants with bone progression). 6. Has received at least 1 line of taxane-based chemotherapy for HSPC or CRPC and have had progressed disease during or on treatment, or refused or ineligible to receive chemotherapy. 7. Has a life expectancy of >3 months.

4.2 Exclusion Criteria 1. Has history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 3 years. 2. Has presence of gastrointestinal condition, e.g. malabsorption, that might affect the adsorption of study intervention. 3. Has a history of pituitary dysfunction. 4. Has poorly controlled diabetes mellitus. 5. Has active or unstable cardio/cerebro-vascular disease, including thromboembolic events. 6. Has undergone major surgery, including local prostate intervention (except prostate biopsy), within 4 weeks of the date of allocation. 7. Has received an anticancer monoclonal antibody (mAb) within 4 weeks of allocation, or has not recovered from adverse events (AEs) due to mAbs administered more than 4 weeks before the date of allocation. 8. Received prior systemic anticancer therapy including investigational agents within 4 weeks before the date of allocation. 9. Has a diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any form of immunosuppressive therapy within 7 days prior to the start of study intervention. 10. Has a known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 11. Has an active autoimmune disease that has required systemic treatment in the past 2 years. 12. Has a history of or current human immunodeficiency virus (HIV) infection. 13. Has a concurrent Hepatitis B or Hepatitis C virus infection. 14. Has a history of allogenic tissue or solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Opevesostat 5 mg oral tablet taken twice daily. It is co-administered with

Dexamethasone (1.5 mg once daily) and Fludrocortisone acetate (0.1 mg once daily). Hydrocortisone is provided as a rescue medication. **Route of Administration:** Oral **Duration of Treatment:** Continuously until unacceptable toxicity or documented disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant administration of oral dexamethasone, fludrocortisone, and hydrocortisone (rescue). **Prohibited:** Recent use of an anticancer monoclonal antibody (mAb) or investigational agents within 4 weeks of allocation. Chronic systemic steroid therapy (exceeding 10 mg daily prednisone equivalent) or immunosuppressive therapy within 7 days prior to intervention is excluded.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Eligibility Criteria Check, Medical History
Baseline	Day 0	First Dose, Safety Baseline	Interim Pre-specified timepoints Pharmacokinetic blood sampling (\$Cmax\$, \$Tmax\$, \$AUC_{0-12}\$, \$t_{1/2}\$), Safety Labs, Tumor Imaging, PSA checks
End of Study	Follow-up	Final vital signs, progression tracking, Survival follow-up	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Number of Participants Who Experience an Adverse Event (AE):** Assessment of safety and tolerability. * **Number of Participants Who Discontinue Study Intervention Due to an AE.** * **Maximum Plasma Concentration (Cmax) of opevesostat:** Blood samples will be collected at pre-specified timepoints to determine the Cmax. Other pharmacokinetic profiling includes parameters such as \$Tmax\$, \$AUC_{0-12}\$, \$t_{1/2}\$, \$CL/F\$, and \$Vz/F\$.

7.2 Secondary & Exploratory Endpoints * **Prostate-specific Antigen (PSA) Response Rate:** Defined as a \$\ge 50\%\$ reduction. * **Radiographic Progression-free Survival (rPFS):** Per Prostate Cancer Working Group (PCWG)-modified Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1). * **Objective Response Rate (ORR):** Percentage of participants with a confirmed Complete Response (CR) or Partial Response (PR) Per PCWG-modified RECIST 1.1. * **Overall Survival (OS) and Duration of Response (DOR).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection continuously throughout the study. An AE is defined as any untoward medical occurrence associated with the use of a drug in a participant,

whether or not considered drug-related. An AE does not imply any judgment about causality. **Serious Adverse Events (SAEs):** Specific monitoring focused on dose-limiting toxicities, discontinuations due to an AE, and clinically significant changes. **Data Monitoring Committee (DMC):** Safety reporting and trial monitoring managed by Merck.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants who received at least one dose of the study drug). **Sample Size Justification:** Target Enrollment = 14 participants; small cohort appropriately scaled for a Phase 1 pharmacokinetics and initial safety characterization. **Handling of Missing Data:** Standard Phase 1 protocols apply for pharmacokinetic concentration data evaluations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency, intensive monitoring to track pharmacokinetics, clinical toxicities, and drug-absorption behaviors in an open-label Phase 1 setup. **Audit Trail:** Continuous clinical record keeping, protocol deviation tracking, and secure tracking of biological PK samples.

11. Study Identifiers & Governance

Primary ID: MK-5684-001 **Registry Number:** NCT06136598 **Posting ID:** 892317133212947536 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** A Study of Opevesostat (MK-5684) in China Participants With Metastatic Castration-Resistant Prostate Cancer (mCRPC) (MK-5684-001) **Official Regulatory Title:** A Phase 1 Clinical Study to Investigate the Safety and Pharmacokinetics of MK-5684 in China Participants With Metastatic Castration-Resistant Prostate Cancer

12. Clinical Framework (Oncology / Prostate Cancer Specific)

Disease Areas: Prostatic Neoplasms, Metastatic Castration-Resistant Prostate Cancer **Primary Condition / Preferred UMLS Name:** Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Radiographic evidence of metastases (bone/soft tissue) and disease progression following NHAs and taxane-based chemotherapy. **Medical Methodology:** Administration of targeted CYP11A1 inhibitor therapy along with synthetic

corticosteroid supplementation (dexamethasone). **Optimization Target:** Identifying the safety profile, maximum tolerated limits, and understanding the pharmacokinetic behaviors of the drug.

13. Study Procedures & Methodology

Management Pathway: Phase 1 Pharmacokinetics and Safety Assessment clinical pathway. **Education Protocol (HCP):** Site training on strict PK sampling schedules, steroid co-administration, and toxicity reporting. **Education Protocol (Patient):** Informed consent review covering experimental drug expectations and reporting of side effects. **Quality Audit Mechanism:** Laboratory assessment of drug absorption, distribution, and overall patient tolerability.

14. Observation & Follow-Up Schedule

Total Duration: From allocation until disease progression, unacceptable toxicity, or death. **Onsite Visit Cadence:** Required at pre-specified timepoints (typically continuous within defined cycles) to obtain necessary blood samples for PK parameters (e.g., C_{max}). **Remote Monitoring Frequency:** Standard protocol AE monitoring. **Checklist Review Interval:** Routine reviews aligned with treatment cycles and tumor progression scan schedules.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					